Combined standard deviation

$$d_1 = \overline{x_{12}} - \overline{x_1} = 4729.166 — 5000 = -270.834$$

$$\sigma_{12} = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

$$= \sqrt{\frac{550(900 + 73,351.05) + 650(1600 + 52,517.05)}{550 + 650}}$$

$$= \sqrt{\frac{4,08,38,080.55 + 3,51,76,082.50}{1200}} = \sqrt{633445} = 251.68$$

6.5 Relative Measures

It is a pure number independent of the units of measurements. This measure is useful especially when the data sets are measured in different units of measurement.

For example, suppose a nutritionist would like to compare the obesity of school children in India and England. He collects data from some of the schools in these two countries. The weight is normally measured in kilograms in India and in pounds in England. It will be meaningless, if we compare the obesity of students using absolute measures. So it is sensible to compare them in relative measures.

6.5.1 Coefficient of Variation

The standard deviation is an absolute measure of dispersion. It is expressed in terms of units in which the original figures are collected and stated. The standard deviation of heights of students cannot be compared with the standard deviation of weights of students, as both are expressed in different units, ie., heights in centimeter and weights in kilograms. Therefore the standard deviation must be converted into a relative measure of dispersion for the purpose of comparison. The relative measure is known as the coefficient of variation.

The coefficient of variation is obtained by dividing the standard deviation by the mean and multiplying it by 100. Symbolically,

$$\text{Coefficient of Variation } (C.V) = \frac{\sigma}{\bar{x}} \times 100$$

If we want to compare the variability of two or more series, we can use *C.V.* The series or groups of data for which the *C.V* is greater indicate that the group is more variable, less stable, less uniform, less consistent or less homogeneous. If the *C.V* is less, it indicates that the group is less variable, more stable, more uniform, more consistent or more homogeneous.

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